

Coordinate Systems and the Celestial Sphere

A Working Guide for Astronomy Olympiads (IOAA / BdOAA)

Mohammad Marshad Bin Khaled

July 22, 2026

Abstract

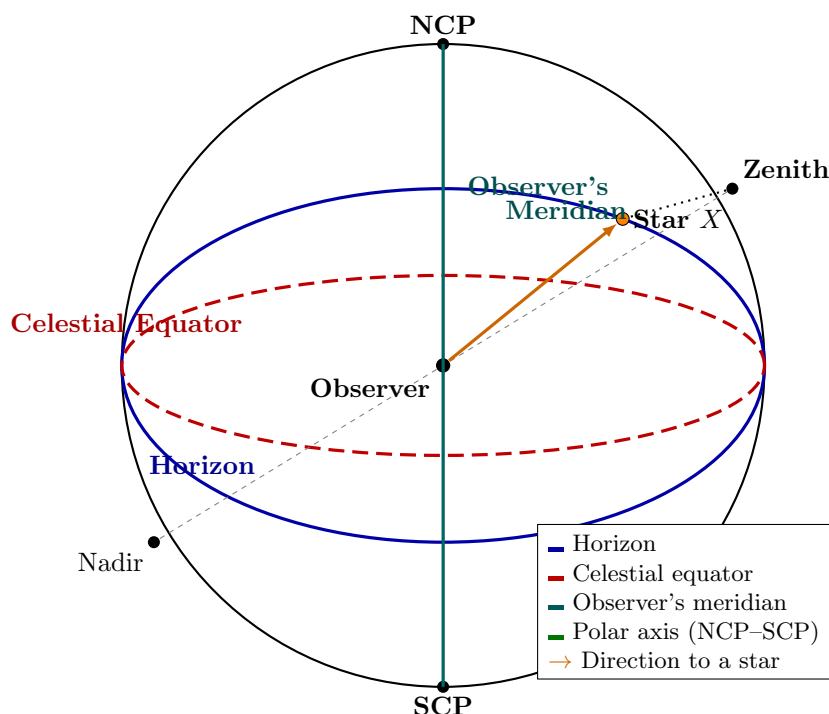
This guide explains the celestial sphere, the four principal astronomical coordinate systems, spherical trigonometry, culmination, circumpolar/never-rising conditions, time-keeping systems, and the sunrise equation — with clearly labelled figures, full derivations, sign conventions, and practice problems in the style of the International Olympiad on Astronomy and Astrophysics (IOAA), IOAA Junior, and the Bangladesh Olympiad on Astronomy and Astrophysics (BdOAA).

Contents

1	The Celestial Sphere	2
2	The Four Coordinate Systems	3
2.1	1. Horizontal (Altitude–Azimuth) System	3
2.2	2. Equatorial System	3
2.3	3. Ecliptic System	4
2.4	4. Galactic System	5
3	The Astronomical Triangle and Spherical Trigonometry	7
3.1	Core Spherical-Trigonometry Formulas	7
4	Transformation Formulas Between the Four Systems	8
4.1	Equatorial $\leftrightarrow$ Horizontal	8
4.2	Equatorial $\leftrightarrow$ Ecliptic	8
4.3	Equatorial $\leftrightarrow$ Galactic	8
5	Culmination: Upper and Lower Transit	10
5.1	The Meridian Diagram	10
5.2	Culmination Altitude Formulas	10
6	Circumpolar and Never-Rising Stars	12
7	Time-Keeping Systems	13
7.1	Key Relations	13
8	The Sunrise Equation	15
8.1	Day Length	15
8.2	Refinement: Refraction and Solar Radius	15
9	Quick Reference Summary	16

1 The Celestial Sphere

The **celestial sphere** is an imaginary sphere of arbitrarily large radius, centred on the observer, onto which all celestial objects are projected. Because the radius is taken to be infinite, only the *direction* to an object matters, never its distance.



Key Points to Remember

- **Zenith:** point directly overhead. **Nadir:** point directly underfoot (opposite the zenith).
- **NCP / SCP:** where Earth's rotation axis, extended infinitely, pierces the sphere.
- **Celestial equator:** Earth's equator projected outward; always 90° from both poles.
- **Horizon:** great circle 90° from the zenith; splits sky into "visible" (above) and "hidden" (below, shaded grey in the figure).
- **Observer's meridian:** the great circle through zenith, nadir, and both celestial poles — this is the circle a star crosses exactly once per day at its *highest* point (see Section 4).

Practice Problems (IOAA / BdOAA style)

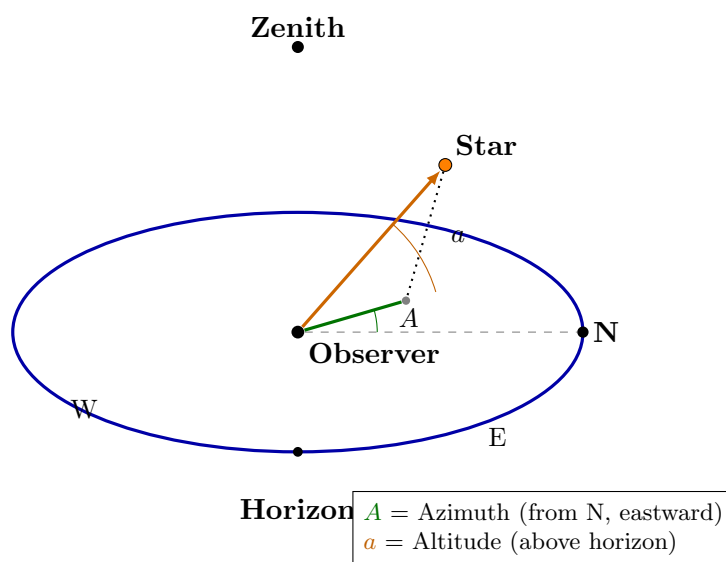
- P1.1** An observer stands at the North Pole of the Earth. Describe, with reasoning, where the zenith, the NCP, and the celestial equator each appear in the sky.
- P1.2** Explain why the celestial equator, the horizon, and the observer's meridian are all *great circles*, while a star's daily path (except for a star exactly on the equator) is *not*.
- P1.3** For an observer at latitude $\phi = 23.5^\circ\text{N}$, what is the angle between the horizon and the celestial equator, measured at the point where they cross due south?

2 The Four Coordinate Systems

Every coordinate system on the sky is fixed by three choices: (i) a **fundamental plane** (its “equator”), (ii) a **pole** perpendicular to that plane, and (iii) a **zero-point** from which the longitude-like angle is counted. This section builds all four systems on exactly that template so they are easy to compare.

2.1 1. Horizontal (Altitude–Azimuth) System

Fundamental plane: the horizon. **Pole:** the zenith. **Zero-point:** the North point.



- **Altitude a :** 0° at the horizon to $+90^\circ$ at the zenith; negative below the horizon. **Zenith distance $z = 90^\circ - a$.**
- **Azimuth A :** measured along the horizon from North, increasing *eastward*, 0° to 360° .

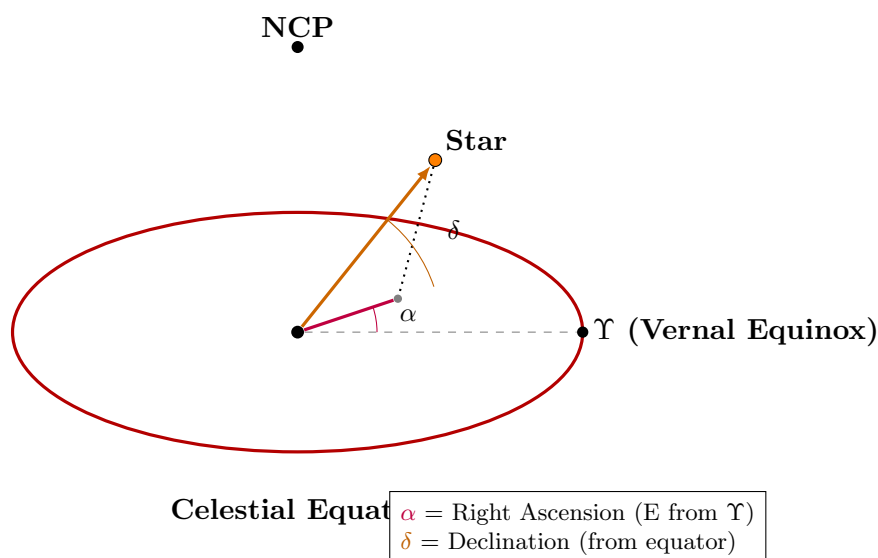
This system is purely local — it depends on the observer’s position and the time of day, and rotates as the sky turns.

Practice Problems (IOAA / BdOAA style)

- P2.1** A star has altitude $a = 0^\circ$ and azimuth $A = 270^\circ$. Where exactly is it, and what is it doing (rising or setting)?
- P2.2** Two stars have the same altitude but azimuths $A_1 = 60^\circ$ and $A_2 = 300^\circ$. Sketch their positions relative to North and comment on any symmetry.
- P2.3** Why can two different stars, observed at two different times, share the exact same (a, A) pair? What does this tell you about the Horizontal system as a way to *catalogue* stars?

2.2 2. Equatorial System

Fundamental plane: the celestial equator. **Pole:** the NCP. **Zero-point:** the Vernal Equinox Υ .



- **Declination δ :** $\pm 90^\circ$ at NCP/SCP; analogous to latitude on Earth.
- **Right Ascension α :** eastward from Υ, usually in hours (0^h – 24^h , $1^h = 15^\circ$).
- **Hour Angle H** (a related but distinct angle): measured *westward* from the observer's meridian instead of from Υ, so $H = \text{LST} - \alpha$.

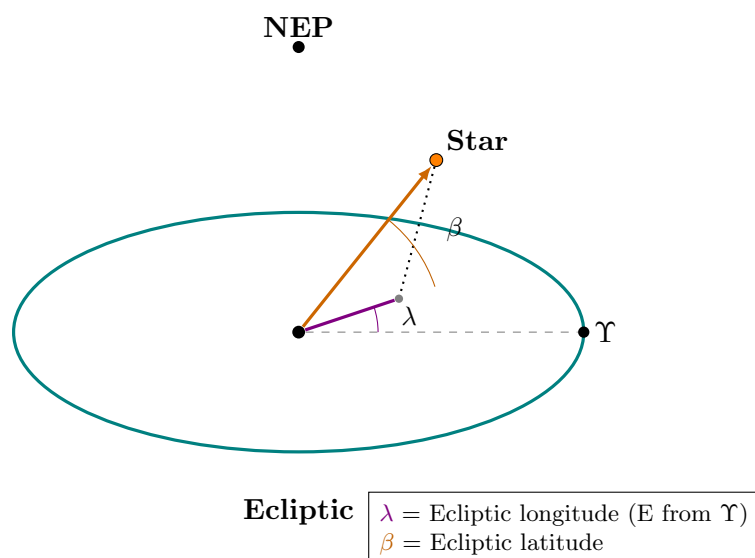
Because it is tied to the stars rather than to the observer, this system *does not* rotate with the local sky — a star's (α, δ) stays essentially fixed year to year.

Practice Problems (IOAA / BdOAA style)

- P3.1** A star has $\alpha = 6^h$, $\delta = +40^\circ$. State, without calculation, whether it is ever circumpolar for an observer at $\phi = +50^\circ\text{N}$ (use the idea, not the formula yet — revisit after Section 5).
- P3.2** If Local Sidereal Time is 14^h30^m and a star has $\alpha = 10^h00^m$, find its hour angle H . Is the star east or west of the meridian?
- P3.3** Explain, in one or two sentences, why Right Ascension is measured in *hours* rather than degrees.

2.3 3. Ecliptic System

Fundamental plane: the ecliptic (Sun's annual path). **Pole:** NEP, tilted $\varepsilon = 23.44^\circ$ from the NCP. **Zero-point:** Υ.



- **Ecliptic latitude β :** $\pm 90^\circ$ at NEP/SEP; nearly 0° for planets (they lie close to the ecliptic).
- **Ecliptic longitude λ :** eastward from Υ , 0° – 360° ; equals the Sun's own λ on the day it is there.

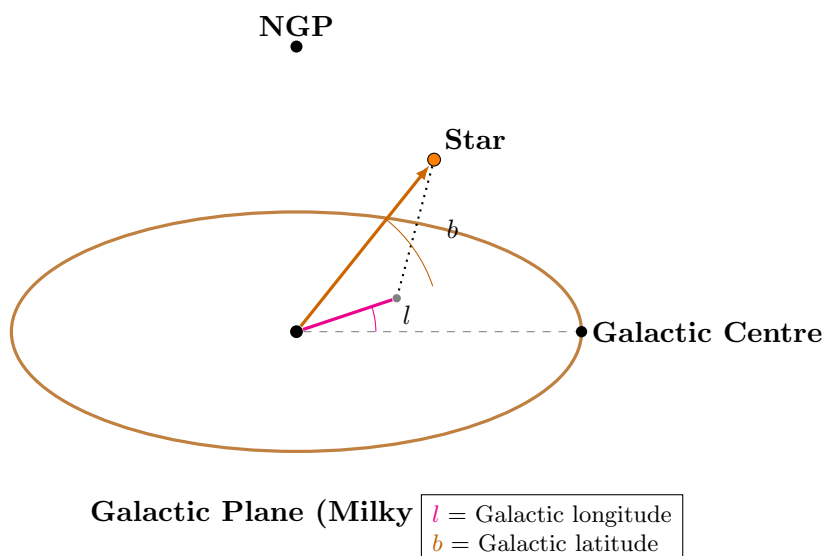
Convenient for Solar-System work (Sun, Moon, planets) since they all stay near $\beta = 0$.

Practice Problems (IOAA / BdOAA style)

- P4.1** On the day of the (northern) summer solstice, what is the Sun's ecliptic longitude $\lambda_\odot$? What is it at the autumnal equinox?
- P4.2** Why does a planet's ecliptic latitude β stay small (typically a few degrees) while its ecliptic longitude λ sweeps through the full 360° over its orbital period?
- P4.3** Using $\varepsilon = 23.44^\circ$, find (α, δ) of the Sun when $\lambda_\odot = 90^\circ$ (hint: use the transformation formulas in Section 4).

2.4 4. Galactic System

Fundamental plane: the Milky Way's disc. **Pole:** NGP ($\alpha_{\text{NGP}} = 192.86^\circ$, $\delta_{\text{NGP}} = 27.13^\circ$, J2000). **Zero-point:** the Galactic Centre.



Used for mapping the Milky Way's structure; a star with $b \approx 0^\circ$ lies in the Galactic disc, while globular clusters and the Galactic halo tend to have large $|b|$.

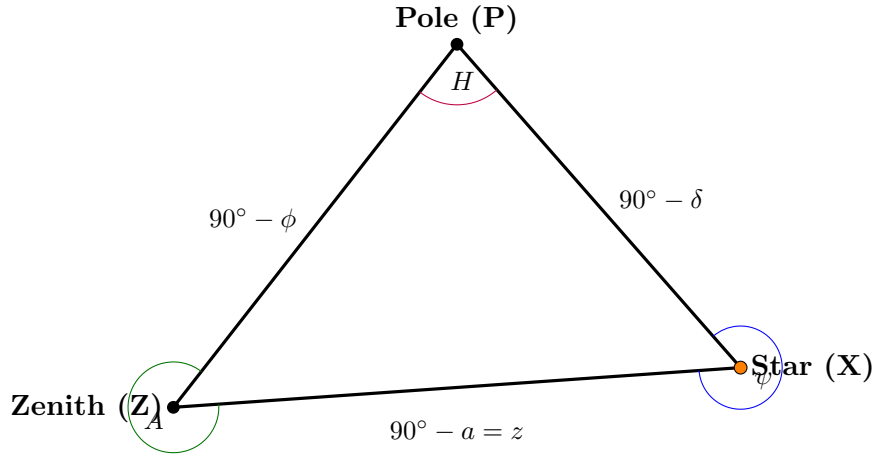
Practice Problems (IOAA / BdOAA style)

P5.1 Why would an astronomer studying the Galaxy's spiral-arm structure prefer Galactic coordinates over Equatorial coordinates?

P5.2 A star has $b = +5^\circ$. Would you expect it to be more or less reddened by interstellar dust than a star with $b = +60^\circ$? Explain briefly.

3 The Astronomical Triangle and Spherical Trigonometry

To convert between systems we use the **Astronomical Triangle**: a spherical triangle with vertices at the **Pole (P)**, the **Zenith (Z)**, and the **Star (X)**.



Sides: $PZ = 90^\circ - \phi$, $PX = 90^\circ - \delta$, $ZX = 90^\circ - a$ Angles: H at P, A at Z, ψ (parallactic) at X

- ϕ = observer's latitude, δ = star's declination, a = star's altitude, z = zenith distance, H = hour angle, A = azimuth, ψ = parallactic angle (how "tilted" the star appears relative to the vertical — important for imaging/spectroscopy).

3.1 Core Spherical-Trigonometry Formulas

For a spherical triangle with sides a, b, c opposite angles A, B, C :

$$\text{Cosine Rule: } \cos a = \cos b \cos c + \sin b \sin c \cos A \quad (1)$$

$$\text{Sine Rule: } \frac{\sin a}{\sin A} = \frac{\sin b}{\sin B} = \frac{\sin c}{\sin C} \quad (2)$$

$$\text{Four-Parts Formula: } \cos a \cos C = \sin a \cot b - \sin C \cot B \quad (3)$$

Applying the cosine rule with sides PZ, PX, ZX and angle H at P gives the Equatorial $\leftrightarrow$ Horizontal transformation of Section 4.

Practice Problems (IOAA / BdOAA style)

P6.1 Derive $\cos z = \sin \phi \sin \delta + \cos \phi \cos \delta \cos H$ starting from the cosine rule applied to the astronomical triangle. State clearly which side plays the role of a, b, c in Eq. (1).

P6.2 Using the sine rule on the astronomical triangle, write down the relation between $\sin A$, $\sin H$, and $\sin z$, δ (used to find azimuth once the altitude is known).

P6.3 A star has $z = 40^\circ$, $\phi = 30^\circ$, $\delta = 20^\circ$. Find its hour angle H .

4 Transformation Formulas Between the Four Systems

4.1 Equatorial $\leftrightarrow$ Horizontal

Given δ, H, ϕ , find a, A :

$$\sin a = \sin \phi \sin \delta + \cos \phi \cos \delta \cos H \quad (4)$$

$$\cos A = \frac{\sin \delta - \sin \phi \sin a}{\cos \phi \cos a} \quad (5)$$

$$\sin A = \frac{-\cos \delta \sin H}{\cos a} \quad (6)$$

Inverse (Horizontal $\rightarrow$ Equatorial):

$$\sin \delta = \sin \phi \sin a + \cos \phi \cos a \cos A \quad (7)$$

$$\cos H = \frac{\sin a - \sin \phi \sin \delta}{\cos \phi \cos \delta} \quad (8)$$

4.2 Equatorial $\leftrightarrow$ Ecliptic

With $\varepsilon = 23.44^\circ$ (obliquity):

$$\sin \beta = \sin \delta \cos \varepsilon - \cos \delta \sin \varepsilon \sin \alpha \quad (9)$$

$$\cos \beta \sin \lambda = \sin \delta \sin \varepsilon + \cos \delta \cos \varepsilon \sin \alpha \quad (10)$$

$$\cos \beta \cos \lambda = \cos \delta \cos \alpha \quad (11)$$

Inverse:

$$\sin \delta = \sin \beta \cos \varepsilon + \cos \beta \sin \varepsilon \sin \lambda \quad (12)$$

$$\cos \delta \sin \alpha = \cos \beta \sin \lambda \cos \varepsilon - \sin \beta \sin \varepsilon \quad (13)$$

$$\cos \delta \cos \alpha = \cos \beta \cos \lambda \quad (14)$$

4.3 Equatorial $\leftrightarrow$ Galactic

With $\alpha_{\text{NGP}} = 192.86^\circ$, $\delta_{\text{NGP}} = 27.13^\circ$, $l_{\text{NCP}} = 122.93^\circ$:

$$\sin b = \sin \delta \sin \delta_{\text{NGP}} + \cos \delta \cos \delta_{\text{NGP}} \cos(\alpha - \alpha_{\text{NGP}}) \quad (15)$$

$$\cos b \sin(l_{\text{NCP}} - l) = \cos \delta \sin(\alpha - \alpha_{\text{NGP}}) \quad (16)$$

$$\cos b \cos(l_{\text{NCP}} - l) = \sin \delta \cos \delta_{\text{NGP}} - \cos \delta \sin \delta_{\text{NGP}} \cos(\alpha - \alpha_{\text{NGP}}) \quad (17)$$

System	Fund. Circle	Pole	Coord. 1	Coord. 2
Horizontal	Horizon	Zenith	Altitude $a \in [-90^\circ, 90^\circ]$	Azimuth $A \in [0^\circ, 360^\circ)$
Equatorial	Celestial equator	NCP	Declination $\delta \in [-90^\circ, 90^\circ]$	RA $\alpha \in [0^h, 24^h)$
Ecliptic	Ecliptic	NEP	Latitude $\beta \in [-90^\circ, 90^\circ]$	Longitude $\lambda \in [0^\circ, 360^\circ)$
Galactic	Galactic plane	NGP	Latitude $b \in [-90^\circ, 90^\circ]$	Longitude $l \in [0^\circ, 360^\circ)$

Key Points to Remember

- All latitude-type coordinates (a, δ, β, b) are + toward their system's North pole.
- All longitude-type coordinates (A, α, λ, l) increase eastward.
- Hour Angle H increases *westward* (opposite α): $H = \text{LST} - \alpha$.

Practice Problems (IOAA / BdOAA style)

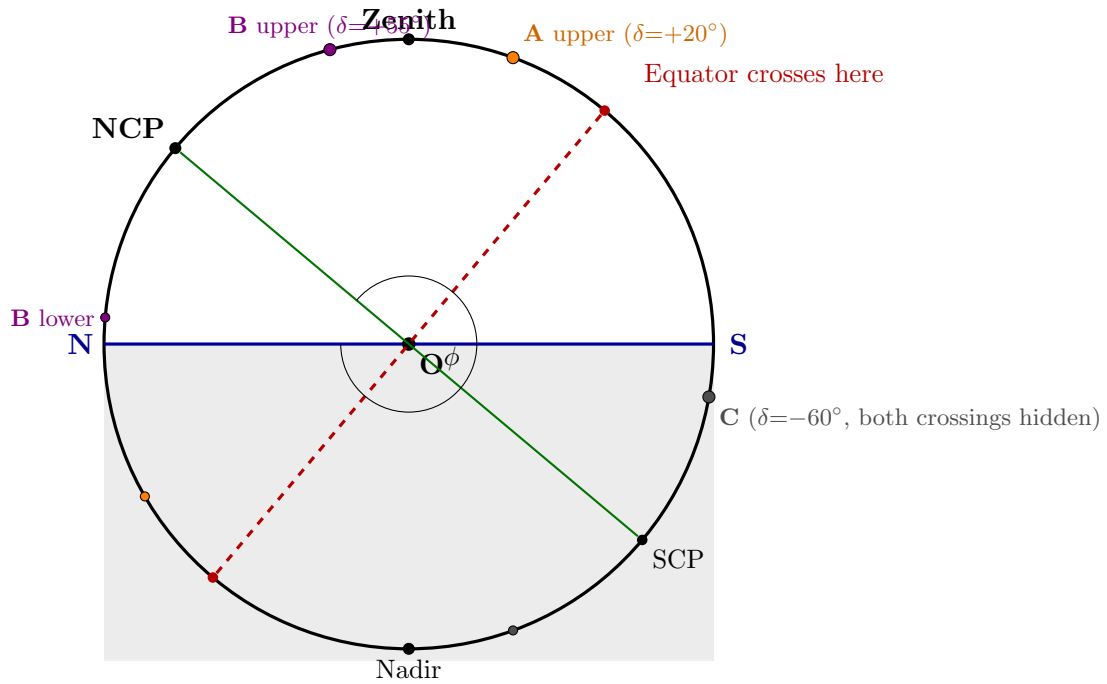
- P7.1** A star is at $\alpha = 5^h, \delta = +30^\circ$. For an observer at $\phi = +40^\circ\text{N}$ when $\text{LST} = 7^h$, find the star's altitude and azimuth.
- P7.2** The Sun has $\lambda_\odot = 0^\circ$ (vernal equinox). Verify using Eqs. (9)–(11) that its $(\alpha, \delta) = (0^h, 0^\circ)$.
- P7.3** Vega has approximately $\alpha = 18^h37^m, \delta = +38.8^\circ$. Estimate its Galactic latitude b (is it near the Galactic plane or far from it?) using Eq. (12).

5 Culmination: Upper and Lower Transit

Culmination (or *transit*) is the moment a star crosses the observer's meridian — the instant it reaches its highest (or, on the far side of the pole, lowest) point in the sky for that day. At culmination the hour angle is $H = 0^h$ (**upper culmination**) or $H = 12^h$ (**lower culmination**).

5.1 The Meridian Diagram

The clearest way to see culmination is a 2-D cross-section of the celestial sphere cut by the observer's meridian plane. All the key points — Zenith, Nadir, both poles, both equator crossings, and a star's two meridian crossings — lie on a single great circle, drawn below for an example observer at latitude $\phi = 40^\circ\text{N}$.



Key Points to Remember

- **Star A** ($\delta = +20^\circ$): both meridian crossings straddle the horizon correctly — upper culmination above, lower culmination below (star rises and sets normally).
- **Star B** ($\delta = +55^\circ$): *both* crossings are above the horizon — the star is **circumpolar** (never sets).
- **Star C** ($\delta = -60^\circ$): *both* crossings are below the horizon (grey region) — the star **never rises** for this observer.

5.2 Culmination Altitude Formulas

$$\text{Upper culmination: } a_{\text{upper}} = 90^\circ - |\phi - \delta| \quad (18)$$

$$\text{Lower culmination: } a_{\text{lower}} = \phi + \delta - 90^\circ \quad (19)$$

(The star is only actually visible at lower culmination if $a_{\text{lower}} \geq 0$, i.e. if it is circumpolar.) At upper culmination the star lies due south of zenith if $\delta < \phi$, and due north of zenith if $\delta > \phi$ (northern-hemisphere observer).

Practice Problems (IOAA / BdOAA style)

- P8.1** For $\phi = 23.5^\circ\text{N}$, find the altitude of the Sun at upper culmination on (a) the summer solstice ($\delta = +23.5^\circ$) and (b) the winter solstice ($\delta = -23.5^\circ$).
- P8.2** A star culminates at altitude 80° , south of the zenith, for an observer at $\phi = 35^\circ\text{N}$. Find its declination.
- P8.3** Show that a star culminates *exactly at the zenith* ($a_{\text{upper}} = 90^\circ$) only when $\delta = \phi$.

6 Circumpolar and Never-Rising Stars

Using the lower-culmination formula, the boundary between “always visible” and “sometimes visible” is set by requiring $a_{\text{lower}} \geq 0$. For a northern-hemisphere observer at latitude $\phi > 0$:

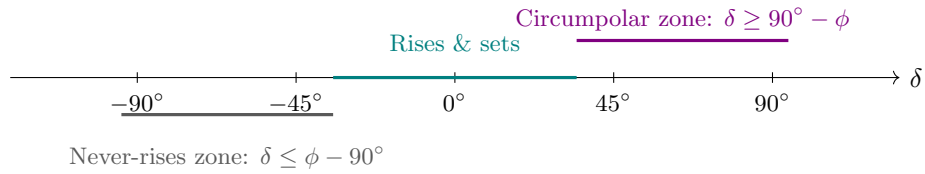
The Three Laws of Circumpolarity

$$\text{Circumpolar (never sets): } \delta \geq 90^\circ - \phi \quad (20)$$

$$\text{Never rises (always hidden): } \delta \leq \phi - 90^\circ \quad (21)$$

$$\text{Rises and sets normally: } \phi - 90^\circ < \delta < 90^\circ - \phi \quad (22)$$

For a southern-hemisphere observer ($\phi < 0$), swap the roles: a star is circumpolar if $\delta \leq -90^\circ - \phi$ (i.e. close to the SCP), by the same symmetry. A convenient hemisphere-independent statement: *a star is circumpolar if it lies within an angular distance $90^\circ - |\phi|$ of the pole on the same side as the observer’s hemisphere; it never rises if it lies within that same distance of the opposite pole.*



Number line sketch for a generic northern observer with $\phi > 0$: the three declination zones (never rises / rises&sets / circumpolar) partition all possible δ .

Practice Problems (IOAA / BdOAA style)

- P9.1** Polaris has $\delta \approx +89.3^\circ$. Below what latitude ϕ does Polaris stop being circumpolar?
- P9.2** For an observer at $\phi = +22^\circ\text{N}$ (roughly Dhaka’s latitude), what range of declinations is circumpolar, and what range never rises?
- P9.3** The star Canopus has $\delta \approx -52.7^\circ$. Can it ever be seen from London ($\phi \approx +51.5^\circ\text{N}$)? Justify using the never-rises law.
- P9.4** A BdOAA-style question: an observer finds that a certain star is exactly *on the boundary* of circumpolarity (it just grazes the horizon at lower culmination). If $\delta = +60^\circ$, find the observer’s latitude.

7 Time-Keeping Systems

Astronomical timekeeping is built on *hour angle*: whichever object defines “noon” or “0^h” for a given system, its hour angle is what the clock measures.

Definitions

- **Local Sidereal Time (LST)**: the hour angle of the Vernal Equinox Υ . Equivalently, LST equals the RA α of whatever star is *at upper culmination right now* (since $H = 0$ there and $H = \text{LST} - \alpha$).
- **Apparent Solar Time**: $H_{\odot} + 12^h$, where $H_{\odot}$ is the true Sun’s hour angle (so that 0^h = local apparent midnight, 12^h = true local noon).
- **Mean Solar Time**: same idea, but using a fictitious “mean Sun” that moves at a perfectly uniform rate along the equator — this is what ordinary clocks approximate.
- **Equation of Time**: $\text{EoT} = \text{Apparent Solar Time} - \text{Mean Solar Time}$, typically ± 16 minutes over the year, caused by (i) the eccentricity of Earth’s orbit and (ii) the obliquity of the ecliptic.
- **Standard (Zone) Time**: Local Mean Time is impractical city-by-city, so a whole time zone adopts the LMT of one reference (standard) meridian. Correction: 4 minutes of time per degree of longitude difference from that standard meridian.
- **Universal Time (UT)**: Mean Solar Time at the Greenwich meridian (0° longitude); historically called GMT.

7.1 Key Relations

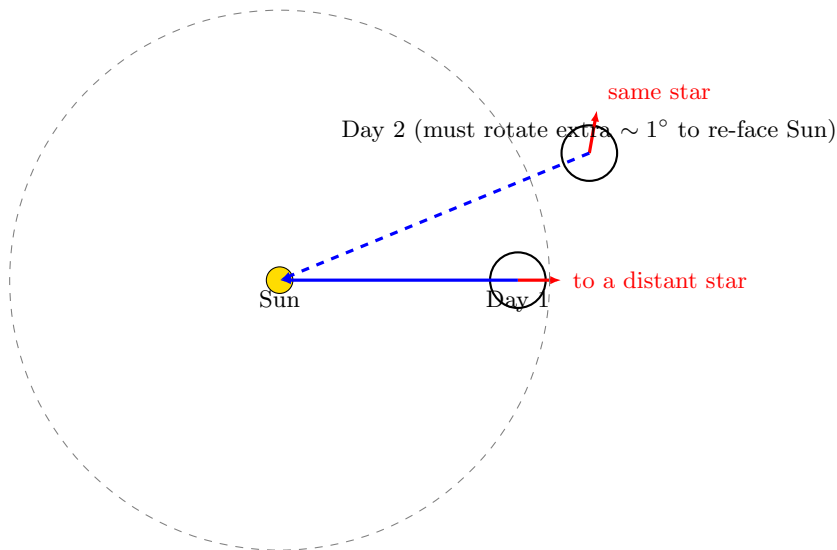
$$\text{LST} = H_{\text{star}} + \alpha_{\text{star}} \quad (\text{for any star, not just at culmination}) \quad (23)$$

$$\text{LST} = \text{GST} + \lambda_{\text{east}} \quad (\lambda_{\text{east}} = \text{observer’s east longitude, in hours}) \quad (24)$$

$$\text{Standard Time} = \text{LMT} - \frac{(\lambda_{\text{local}} - \lambda_{\text{standard}})}{15^\circ/\text{h}} \quad (\lambda \text{ east-positive, in degrees}) \quad (25)$$

$$\text{Sidereal day} = 23^h 56^m 4.09^s \quad \text{Solar (mean) day} = 24^h 00^m 00^s \quad (26)$$

The sidereal day is ~ 4 minutes shorter than the solar day because, while the Earth rotates once on its axis, it has also advanced $\sim 1^\circ$ along its orbit, so it must rotate a little *extra* for the Sun to return to the same meridian.



Practice Problems (IOAA / BdOAA style)

- P10.1** If GST (Greenwich Sidereal Time) is 3^h20^m and an observer is at east longitude 45° , find their LST.
- P10.2** A star culminates (upper) at Local Mean Time 23^h47^m . What is its Right Ascension, expressed as LST at that moment (assume $\text{LST} = \text{LMT} + \text{a known offset}$ — state what extra information you would need to convert exactly)?
- P10.3** On a day when the Equation of Time is +14 minutes, the mean Sun transits (crosses the meridian) at 12:00:00 Mean Time. At what Mean Time does the true (apparent) Sun actually transit?
- P10.4** Bangladesh Standard Time is fixed to the meridian 90°E . Find the Standard-Time correction (in minutes) for a city at longitude 88°E .

8 The Sunrise Equation

At sunrise or sunset the Sun's altitude is (to first approximation) $a = 0^\circ$. Setting $a = 0$ in the altitude formula of Section 4 and solving for H gives the **sunrise equation**:

$$\boxed{\cos H_0 = -\tan \phi \tan \delta} \quad (27)$$

where H_0 is the Sun's hour angle at sunrise/sunset (a negative H_0 before noon = sunrise, positive = sunset, by symmetry $|H_{\text{sunrise}}| = |H_{\text{sunset}}| = H_0$).

8.1 Day Length

Since the Sun moves 15° per hour in hour angle, and it is above the horizon for a total hour-angle range of $2H_0$:

$$\text{Day length} = \frac{2H_0}{15^\circ} \text{ hours} \quad (H_0 \text{ in degrees}) \quad (28)$$

8.2 Refinement: Refraction and Solar Radius

Real sunrise/sunset is defined by the Sun's *upper limb* touching the horizon, not its centre, and light bends slightly through the atmosphere. Both effects are folded into a small negative altitude $a_0 \approx -0.833^\circ$ ($\approx -50'$, combining $\approx -34'$ refraction and $\approx -16'$ solar semi-diameter):

$$\cos H_0 = \frac{\sin a_0 - \sin \phi \sin \delta}{\cos \phi \cos \delta}, \quad a_0 = -0.833^\circ \quad (29)$$

Special Cases

- **Equinox** ($\delta = 0$): $\cos H_0 = 0 \Rightarrow H_0 = 90^\circ$, i.e. a 12-hour day everywhere on Earth (ignoring refraction).
- **Midnight Sun**: if $|\tan \phi \tan \delta| > 1$ with ϕ, δ the *same* sign, Eq. (18) has no solution because the Sun never sets — this happens inside the polar circle in local summer.
- **Polar Night**: same condition but ϕ, δ of *opposite* sign — the Sun never rises.

Practice Problems (IOAA / BdOAA style)

- P11.1** Find the day length at $\phi = 40^\circ\text{N}$ on the day the Sun's declination is $\delta = +15^\circ$ (ignore refraction).
- P11.2** Show that at the Arctic Circle ($\phi = 66.5^\circ\text{N}$), the summer solstice ($\delta = +23.5^\circ$) gives $\cos H_0 = -1$ exactly, i.e. the Sun grazes the horizon at local midnight without setting.
- P11.3** Estimate, using Eq. (19), the sunrise hour angle at $\phi = 23.5^\circ\text{N}$ when $\delta = 0^\circ$, including the $a_0 = -0.833^\circ$ correction — by how many minutes does daylight exceed the simple 12-hour equinox estimate?
- P11.4** An observer at $\phi = 80^\circ\text{N}$ wants to know the last day of the year the Sun rises at all before polar night begins. Set up (you need not fully solve) the inequality in δ that marks this boundary.

9 Quick Reference Summary

Quantity	Formula
Altitude from H, δ, ϕ	$\sin a = \sin \phi \sin \delta + \cos \phi \cos \delta \cos H$
Upper culmination altitude	$a_{\text{upper}} = 90^\circ - \phi - \delta $
Lower culmination altitude	$a_{\text{lower}} = \phi + \delta - 90^\circ$
Circumpolar condition ($\phi > 0$)	$\delta \geq 90^\circ - \phi$
Never-rises condition ($\phi > 0$)	$\delta \leq \phi - 90^\circ$
Sunrise/sunset hour angle	$\cos H_0 = -\tan \phi \tan \delta$
Day length	$2H_0/15^\circ$ hours
LST–Hour Angle relation	$\text{LST} = H + \alpha$
Sidereal vs. solar day	$23^h 56^m 4^s$ vs. $24^h 00^m 00^s$

Compiled as an olympiad-preparation reference by Mohammad Marshad Bin Khaled. Diagrams are schematic and use an example latitude of $\phi = 40^\circ$ purely for illustration.